

Yuvraj Singh

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Full-stack engineer (Node.js • TypeScript • React • MongoDB • PostgreSQL) who ships using **Claude Code and Cursor as primary workflow tools** — not as toys. Comfortable decomposing specs into parallel agent workstreams, verifying output, catching confidently wrong generations, and owning what ships. Available to work from **Jaipur office**.

HOW I WORK WITH AI AGENTS

- Run **parallel Claude Code workstreams** against written specs: decompose into isolated tasks, dispatch concurrently, diff and review all agent output before merging
- Catch agent failure modes — hallucinated APIs, plausible-but-wrong logic, missing edge cases, insecure defaults — and override or reroute rather than accept blindly
- Have thrown away large AI-generated codebases when the abstraction was wrong; restarted from a corrected spec rather than patching bad foundations
- Use Claude, Cursor, and GitHub Copilot daily across frontend, backend, schema design, and test generation — the workflow, not the keystrokes, is what I optimise

EXPERIENCE

AI Research Engineer — Quantum Computing Lab

2024

Bennett University

- Reduced key error rate by **18%** by decomposing simulation spec into isolated Python/Qiskit modules, running parallel implementations, and selecting the lowest-error variant
- Accelerated protocol analysis by **3x** by generating and iterating Matplotlib/Seaborn dashboard code with Claude, verifying output against known quantum noise benchmarks

PROJECTS

NexEco – Smart Travel Buddy

2024

React.js • Next.js • TypeScript • Node.js • MongoDB • PostgreSQL • Docker • Redis • GitHub Actions

- Shipped full-stack travel app by **decomposing product spec into 6 parallel agent workstreams** — schema design, API layer, auth, frontend components, CI/CD, and Docker config — then integrating and owning the result
- Reduced MongoDB retrieval latency by **30%** by reviewing agent-generated schema, identifying missing indexes, and correcting before deployment
- Caught agent-generated GraphQL resolvers with N+1 query bugs; rewrote with DataLoader batching, reducing over-fetching by **35%**
- Configured GitHub Actions CI/CD with Jest test gates; flagged and fixed 3 agent-generated test files that passed on mocks but missed real async edge cases

Customer Churn Prediction — ML Pipeline

2025–2026

Python • FastAPI • Scikit-learn • Pandas • PostgreSQL • NLP

- Built end-to-end ML pipeline achieving **87% ROC-AUC**; used Claude to generate feature engineering code, verified statistical correctness by hand before training
- Cut false-negative predictions by **22%** by iterating threshold logic with agent assistance while manually validating confusion matrix outputs at each step

TECHNICAL SKILLS

Stack	Node.js, TypeScript, React.js, Next.js, MongoDB, PostgreSQL, Redis, Docker, AWS
AI Tools	Claude Code, Cursor, GitHub Copilot — daily workflow, parallel workstreams
Familiar	FastAPI, Python, Jest, GitHub Actions, GraphQL, REST APIs, Figma
Languages	TypeScript, JavaScript, Python, C++, SQL

EDUCATION

B.Tech in Computer Science Engineering

Aug 2023 – Aug 2027

Bennett University, Greater Noida Relevant coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Software Engineering

Higher Secondary (Class XII) & Secondary (Class X) — CBSE

2020 – 2022

MGM Higher Secondary School, Raipur

CERTIFICATIONS

Develop Solutions for Microsoft Azure — Microsoft

Jun 2025

Cloud deployment, API integration, and scalable application design on Azure

Getting Started with Accelerated Computing in CUDA C++ — NVIDIA

Dec 2025

GPU programming, CUDA kernels, memory hierarchy, and parallel computing concepts